

Summary

Parts to the project needed:

1. CAD/Physical Hardware
2. (Digital) Electronics
3. Software

How to execute each + itemized estimated BOM:

1. CAD

- Soldering Kit (maybe?)
- 6x6 Cube w/lid (attached via screws)

2. Electronics

- ESP32
- Perma-Proto/Perfboard
- Jumper Cables
- Pin Headers
- 5v Battery (Small power bank)
- (6) NeoPixel RGBW Mini Button PCB - Pack of 10

3. Software

- Main Code:
 - Rubik's Cube Shuffles (Algorithms/Databases)
 - Address each LED separately (1 - 54)
 - Once affected LEDs get signal, switch orders of LEDs in row (move around)
 - Display new rotation
- Controller Code:
 - Note User Input
 - Show 3D simulation? Simple 'choose face' 'choose row' 'choose movement'?
 - Send to main code
 - Apply to affected pieces

BOM

Part Name	Quantity	Purpose	Source (Company)	Cost	Shipping	Total
CAD	1x	Shell	JLCPCB	Estimate: \$30	\$0	\$30
Soldering Kit	1x	Soldering	Amazon	\$10	\$0 (PRIME)	\$10
Screws	4x	Shell Lid	Home Depot(?)	\$3	\$0 (in person)	\$3
ESP32	1x	Microcontroller	Amazon	\$9.5	\$0 (PRIME)	\$9.5
Perfboard	1x	Homemade PCBboard (time constraint)	AliExpress	\$1.53	\$0 (New Discount)	\$1.53
Perma-Proto Board	1x	Simpler Homemade PCBboard	Adafruit	\$5	\$0 (w/Neo)	\$5
MtoM Jumper Cables	A lot	Connecting (via soldering) components	AliExpress	\$1.53	\$0 (New Discount)	\$1.53
Pin Headers	A lot	Connecting (via soldering) components to board (ex: ESP32)	AliExpress	\$0.73	\$0 (New Discount)	\$0.73
(5v) BatteryBank	1x	Used to power the Neopixels	Amazon	\$10	\$0 (prime)	\$10
(3.3v) BatteryLipo	1x	Used to power the ESP32	Amazon	\$7	\$0 (prime)	\$7
Neopixels 10 pck	6x	LED choice for RubiX Cube	Adafruit	\$29.7	\$18.36 w/others	\$48.06
Acrylic Sheets	6x	Provides a nice display finish	Friend :D	\$3-5	\$0 (Outpost)	\$3-5
Totals						126.35

List of things to do:

~~Resubmit blinky~~

~~Write notes on LED setup (circuits)~~

~~Find addressable leds i could use~~

Create CAD asap

Make the breadboard via a virtual breadboard app (½ done)

- Ive run into a few issues here:
 - Battery 5v vs 3.3v
 - To elaborate here, 5v is required for the neopixels; however, the esp32 only takes 3.3v (2.6v - 3.6v)
 - This issue is also elaborated on by this guy: ede
 - However, there's a fix (?)
 - A logic level shifter (???)
<https://www.youtube.com/watch?v=y6HiDnyhRa0>
 - Yeaah i need a bidirectional level shifter... saved in BOM tab
 - Also, new info: JLCPCB HAS 3D PRINTING THAT SHIPS IN 2-4 DAYS??? This could be goats
- Perfboard to make a 'pcb'?

Adjust CAD if necessary

Submit design!

Research timelapse! So far ive come up with some prompts to ask ai how i can approach the project + asked some people + did some material research and such

i want to make a rubix cube that uses leds to change colors as in to 'shuffle' (ill probably use a database to combine combinations) and then you can still turn it to solve it, how would i even approach that? hardware parts, kicad, cad, sizing, etc... a plan? what components i might need?

i want to make a rubix cube that uses leds to change colors as in to 'shuffle' (ill probably use a database to combine combinations) and then you can still turn it to solve it, how would i even approach that? hardware parts, kicad, cad, sizing, etc... a plan? what components i might need?

i was basically thinking to get twelve mini leds that i could wire to a certain number (54 leds; they each go on the outside of the cube, or are kinda drilled into the case) then there are some components in the middle (the cube will be transparent btw preordered) then i can just code it however?

i want to make a rubix cube that uses leds to change colors as in to 'shuffle' (ill probably use a database to combine combinations) and then you can still turn it to solve it, send approach? hardware parts, kicad, cad, sizing, etc... a plan? what components i might need? (54 leds; they each go on the outside of the cube, or are kinda drilled into the case) then there are some components in the middle (the cube will be transparent btw preordered) i think it would have to be wireless.. <https://www.adafruit.com/product/5140> i looked at this led product and i think it might work

okay so basically its like a cube but w/leds instead of a turning mechanism (kinda like a rubiks cube) you swipe up on a separate controller (probs gonna make a web app for the controller) and it responds by moving up the lights as if it was turning so 54 leds

(probs addressable leds), i need a battery, rubiks cube cad, but basically as for the pcb or mcu i need it to basically use the input from the software controller, transfer it to the code (each led will have an assigned number maybe? someone was saying to use an expander to reduce the amount of pins i need for the leds) and then the output of the code would be to change the led position

New idea: im gonna use 6 square lcd displays that are also touchscreen; they'll be fitted onto a cube and probably use a cube battery

https://www.aliexpress.us/item/3256808943417553.html?spm=a2g0o.productlist.main.6.c991Cx7OCx7Ojy&algo_pvid=21ec8764-cfe1-4d00-9f20-61e4d4f1daea&algo_exp_id=21ec8764-cfe1-4d00-9f20-61e4d4f1daea-5&pdp_ext_f=%7B%22order%22%3A%2215%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21USD%217.75%217.75%21%21%217.75%217.75%21%402101eecd17822238392008106e6428%2112000048018932887%21sea%21US%210%21ABX%211%210%21n_tag%3A-29910%3Bd%3A41b51481%3Bm03_new_user%3A-29895&curPageLogUid=yollNP5SQNTX&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005009129732305%7C_p_origin_prod%3A

I need to build a case for this ^ w/dimensions

And probably first, pcb things

Research will probably be done via chatgpt or ai 🤖

Confirmed BOM (mostly):

6/7x -

https://www.aliexpress.us/item/3256808943417553.html?spm=a2g0o.detail.0.0.45adNynRNynRhF&productId=3256808943417553&pdp_ext_f=%7B%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%2C%22tabScene%22%3A%22wholesale%22%2C%22sku_id%22%3A12000048018932888%2C%22origProductId%22%3A%223256808943417553%22%2C%22order%22%3A%2215%22%7D

Raspberry pi(?):

What do i need

Cube structure

Leds

<https://www.adafruit.com/product/5352>

Wireless charger

https://www.reddit.com/r/esp32/comments/lqi5x9/can_the_esp32_control_6_separate_ws2812_led_strips/ <https://www.willdonaldson.io/blog/rubiks-cube-with-no-moving-parts-part1>

<https://www.instructables.com/Wireless-LED-Cube-Light/>

<https://link.springer.com/article/10.1186/s10033-020-00500-z>

Material list:

https://www.aliexpress.us/item/3256812016867800.html?spm=a2g0o.productlist.main.1.4b53SrW8SrW8HB&algo_pvid=2e48f307-ba86-44ea-a42a-00aeb8532e95&algo_exp_id=2e48f307-ba86-44ea-a42a-00aeb8532e95-0&pdp_ext_f=%7B%22order%22%3A%22-1%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21USD%2126.00%2112.22%21%21%21174.76%2182.14%21%402101eee917816729266647981e705

[4%2112000057738009577%21sea%21US%210%21ABX%211%210%21n_tag%3A-29910%3Bd%3A41b51481%3Bm03_new_user%3A-29895&curPageLogUid=YFYIbQkG6iQs&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005012203182552%7C_p_origin_prod%3A](https://www.google.com/search?q=WS2812b+with+arduino+uno&rlz=1CAKNUI_enUS1181&oq=WS2812b+with+arduino+uno&gs_lcrp=EgZjaHJvbWUyBggAEEUYOTIHCAEQIRiPAjIHCAIQIRiPAIIBCDQyODRqMGo3qAIAAsAIA&sourceid=chrome&ie=UTF-8&surl=1&safe=active&ssui=on)
<https://learn.illuminations.mit.edu/chapter/ws2812b>

Tutorials:

https://www.google.com/search?q=WS2812b+with+arduino+uno&rlz=1CAKNUI_enUS1181&oq=WS2812b+with+arduino+uno&gs_lcrp=EgZjaHJvbWUyBggAEEUYOTIHCAEQIRiPAjIHCAIQIRiPAIIBCDQyODRqMGo3qAIAAsAIA&sourceid=chrome&ie=UTF-8&surl=1&safe=active&ssui=on
<https://www.youtube.com/watch?v=nrM7G36ldpQ>
https://www.flux.ai/p/blog/pcb-design-what-you-need-to-know-before-you-start?utm_source=google&utm_medium=ppc&utm_campaign=gs_ppc_t1_pmax_trial_all-devices_tcpa_multi_eq_all-audiences-us&utm_content=all_multi-targeting_homepage_pmax&utm_term=_&gad_source=1&gad_campaignid=23338870933&gbraid=0AAAAABS1H_9cbarPq53ZAOPhhu4mtJ9DQ&gclid=Cj0KCQjwxvjRBhC2ARIsAI7KJa0mSTXfY45J5uYH1d86s6jx_MppRuS3zUkmabl1LjcnYqY66KTukOQaAif5EALw_wcB

54 cubes = 54 leds